

Expression of Interest ATR Ion Exchange Column Fabrication

1. INTRODUCTION

Battelle Energy Alliance, LLC (BEA) Management & Operating Contractor of the U.S. Department of Energy (DOE) owned Idaho National Laboratory (INL), is seeking an Expression of Interest (EOI) from qualified vendors capable of fabricating a complete Ion Exchange Column Assembly to support operations at the Advanced Test Reactor (ATR). The fabricated system will play a critical role in radiochemical process performance, shielding effectiveness, and facility reliability.

2. BACKGROUND

INL is a U.S. Department of Energy (DOE) national laboratory dedicated to nuclear energy, national security, and clean energy innovation. The ATR facility relies on highly specialized purification and shielding systems, which include the Ion Exchange Column Assembly. BEA is requesting vendor capabilities statements for the fabrication of a multi-component assembly that must meet strict nuclear, metallurgical, and structural performance standards. Project-specific INL engineering reports, detailed drawings, structural analyses, and procedural references will be furnished.

The scope of work includes the following:

- Fabrication of one complete Ion Exchange Column Assembly, including:
 - Two stainless steel ion exchange columns
 - Shield box
 - Access port covers
 - Lead shot shielding material
 - All mounting hardware and handling hardware following detailed drawings and engineering reports
- Development of procedures for lead-shot and rigging handling of the completed assembly

The scope of work **does not** include the following:

- Installation at the facility
- Interfacing equipment beyond supplied mounting hardware
- Overpressure protection of the final assembly

3. APPLICABLE CODES AND REFERENCES

The fabrication and associated documentation must comply with the following (as applicable to component type):

- ASME Section VIII Division 1(2023)
- ACI 349-13 (Reinforced Concrete Safety-Related Structures)
- ANSI/AISC N690-18 (Nuclear Facility Steel Structures)
- ASCE 4-16 and ASCE 43-19 (Seismic Design)
- ANS-2.26 (Seismic Categorization)

4. DESIGN REQUIREMENTS

Vendors should have demonstrated experience with nuclear-grade fabrication and must meet or exceed the following design conditions:

- Materials & Components:
 - Stainless steel process vessels
 - Lead-shot-filled shielding support frame
 - Structural framing and supports per AISC nuclear standards
- Pressure & Temperature Ratings:
 - Design Pressure: 2500 psig
 - Temperature Range: -20°F to 650°F
- Radiation Shielding:
 - Assembly must reduce radiation fields to below 100 mRem/hr at 30 cm
- Seismic Requirements:
 - Seismic Design Category: SDC-4
 - Design consistent with ASCE 4-16, ASCE 43-19, ANS-2.26
- Inspections & Engineering Approval:
 - All pressure boundary welds inspected per ASME VIII/NDE requirements
 - Fabrication drawings and calculations must be submitted for formal BEA review/approval

5. PROCESS

This EOI is the first step in identifying interested and qualified vendors. Responses may be used to inform the development of a formal solicitation list. Selected vendors may be invited to participate in a Request for Proposal (RFP) process for award consideration.

6. SUBMITTAL REQUIREMENTS

Interested vendors should submit an Expression of Interest that includes the following:

- Company name and point of contact

- Relevant fabrication experience, particularly nuclear or DOE-grade work
- Applicable NAICS codes and business size classification
- Any existing DOE or federal contracting experience (if applicable)
- Demonstrated capability to meet ASME, AISC, seismic, and shielding requirements
- Acknowledgment of the ability to meet the following vendor data submittals:
 - Fabrication drawings and bill of materials
 - Design calculations (structural, seismic, pressure)
 - Material test reports (MTRs)
 - Weld maps and WPS/PQR documentation
 - NDE reports and inspection records
 - Hydrostatic test procedures and final test reports
 - Packaging, handling, and lead-shot management procedures
 - Software verification documentation (if engineering software is used)
 - Completed Vendor Data Schedule and transmittal forms

7. QUALITY ASSURANCE REQUIREMENTS

The subcontractor's quality program must be equivalent to ASME NQA-1. Requirements include:

- Qualification records for welders, inspectors, technicians
- Third-party inspection where required
- Pressure vessel registration (if applicable)
- Certificates of Conformance (CoC)
- Submittal of all quality-related inspection and test documents

8. FABRICATION & TESTING

- Materials must meet applicable ASME and ASTM specifications
- Fabrication must follow ASME Section VIII and ANSI/AISC N690-18 for structural parts
- All surfaces cleaned and finished uniformly; stainless surfaces must meet chloride and contamination requirements
- Identification tags and ASME nameplates applied as required
- Hydrostatic testing must meet ASME VIII requirements, using low-chloride test fluid
- BEA inspectors may witness fabrication and testing activities
- A "U" stamp certification of the vessels

9. DELIVERY & PROTECTION

Final assembly must be delivered to a specified Idaho Falls logistics facility with:

- Cleanliness requirements met; no fluids or contaminants left inside components
- Protective packaging to prevent component damage
- Clear labeling and lift/rigging points identified
- Unit prepared for forklift or crane handling

ATR Ion Exchange Column Fabrication Expression of Interest (EOI)



- INL personnel retain rights for inspections, audits, and hold-point reviews throughout fabrication. Final acceptance occurs after successful hydrostatic testing, inspection, and delivery to an Idaho Falls logistics facility

Deadline for submission: April 22, 2026

Submit to: Chase.Egbert@inl.gov

Only select respondents, as determined by BEA, will receive an invitation to participate in the anticipated RFP solicitation.

10. PROPRIETARY INFORMATION

Because information received in response to this EOI may be business sensitive or proprietary to the responding party, respondents must mark information in their responses accordingly.